

DSA Notes - Mathematical Derivation and Search Space Reduction Pattern

Qsn: — Represent a Number as Sum of Consecutive Natural Numbers

Problem

Given an integer n , count the number of ways to represent it as the sum of **2 or more consecutive natural numbers**.

Examples

```
Input : 10
Output: 1
```

```
10 = 1+2+3+4
```

```
Input : 15
Output: 3
```

```
15 = 1+2+3+4+5
```

```
15 = 4+5+6
```

```
15 = 7+8
```

Constraints

```
1 <= n <= 10^8
```

Brute force should not be $O(n^2)$.

First Observation

Question contains

Consecutive Numbers

Whenever you see

- consecutive integers
- consecutive natural numbers
- consecutive positive numbers

Think

Arithmetic Progression (AP)

How to Think

Suppose one answer is

$4+5+6$

It has only two unknowns.

Starting Number
Length

Let's call them

start = a
length = k

Example

$7+8$

a=7

k=2

Example

$$4+5+6$$

$$a=4$$

$$k=3$$

Example

$$1+2+3+4+5$$

$$a=1$$

$$k=5$$

Every possible answer can be represented using

$$(a, k)$$

This is the biggest observation.

Brute Force

Idea

Try every starting point.

$$1$$

$$1+2$$

$$1+2+3$$

$$1+2+3+4$$

...

Then

$$2$$

$$2+3$$

$$2+3+4$$

...

Continue until

start=n

Pseudo Code

```
count=0

for start in range(1,n):

    total=0

    for x in range(start,n):

        total+=x

        if total==n:
            count+=1

        elif total>n:
            break
```

Time

Worst Case

$O(n^2)$

Too slow.

AP Formula

Sequence

a

a+1

a+2

...

$a+k-1$

Sum of AP

$$\text{Sum} = k(\text{First}+\text{Last})/2$$

First

a

Last

$$a+k-1$$

Therefore

$$\begin{aligned}\text{Sum} \\ &= \\ k(2a+k-1)/2\end{aligned}$$

Since $\text{Sum}=n$

$$n=k(2a+k-1)/2$$

This is the main equation.

Logic Derivation

Multiply by 2

$$2n=k(2a+k-1)$$

Divide by k

$$2n/k = 2a + k - 1$$

Move terms

$$2a = 2n/k - k + 1$$

Finally

$$a = (2n/k - k + 1) / 2$$

Now notice something important.

Instead of trying every

a

we can try every

k

and directly calculate

a

Huge optimization.

Why Iterate over k instead of a?

Possible values of

a

1...

n

Almost

10^8

Too many.

Now think about

k

Can

$k=100000$

ever work for

$n=15$

No.

Because the smallest possible sequence of length 100000 is

$1+2+3+\dots$

already enormous.

So k has a much smaller search space.

How did $\sqrt{2n}$ come?

Smallest possible sum for length k

$1+2+3+\dots+k$

Formula

$$k(k+1)/2$$

This must be $\leq n$.

$$k(k+1)/2 \leq n$$

Multiply by 2

$$k(k+1) \leq 2n$$

Approximately

$$k^2 \leq 2n$$

Hence

$$k \leq \sqrt{2n}$$

This becomes our loop limit.

Instead of

1...

100000000

we check only

1...

14142

for $n=10^8$.

Edge Conditions

We derived

$$a = (2n/k - k + 1) / 2$$

Now ask three questions.

Condition 1

Can

$$2n$$

be divided by

$$k$$

exactly?

Example

$$30/4$$

$$=$$

$$7.5$$

Impossible.

So

```
if (2*n)%k==0
```

Otherwise reject immediately.

Condition 2

After computing

$$x = 2n/k - k + 1$$

Remember

$$2a=x$$

If

$$x=8$$

$$a=4$$

Good.

If

$$x=9$$

$$a=4.5$$

Impossible.

Therefore

$$x \% 2 == 0$$

Condition 3

Need

$$a \geq 1$$

Natural numbers cannot start from

$$0$$

$$-1$$

-2

Since

$$2a=x$$

Simply check

$$x > 0$$

Complete Algorithm

For every possible length

↓

Check divisibility

↓

Compute x

↓

Check x is even

↓

Check x positive

↓

Answer++

Dry Run

$$n=15$$

$$2n=30$$

Maximum

$$\sqrt{30}\approx 5$$

Loop

$$k=2$$

$$30\%2==0$$

Yes

$$x=30//2-2+1$$

$$=14$$

Even

$$a=7$$

Answer

$$7+8$$

$$k=3$$

$$30\%3==0$$

$$x=10-3+1$$

$$=8$$

Even

$$a=4$$

Answer

$$4+5+6$$

k=4

$$30\%4!=0$$

Reject

k=5

$$30\%5==0$$

$$x=6-5+1$$

$$=2$$

Even

$$a=1$$

Answer

$$1+2+3+4+5$$

Count

$$3$$

Correct.

Optimized Code

```
import math

def countWays(n):

    ans = 0

    # Try every possible length
    for k in range(2, int(math.sqrt(2*n)) + 1):

        # 2n must be divisible by k
        if (2*n) % k == 0:

            # x = 2a
            x = (2*n)//k - k + 1

            # x must be positive and even
            if x > 0 and x % 2 == 0:
                ans += 1

    return ans
```

Code Building Line by Line

```
ans=0
```

Store total answers.

```
for k in range(2,int(math.sqrt(2*n))+1):
```

Try every possible sequence length.

Starts from

```
2
```

because question asks

2 or more numbers.

Ends at

$\sqrt{2n}$

because longer lengths are impossible.

```
if (2*n)%k==0
```

Need

$2n/k$

to be an integer.

```
x=(2*n)//k-k+1
```

This is

$2a$

derived mathematically.

```
if x>0 and x%2==0
```

Need

$a>0$

Need

```
a
```

to be integer.

```
ans+=1
```

Found one valid sequence.

Complexity

Time

```
O(√n)
```

Space

```
O(1)
```

Pattern Learned Today

Whenever question contains

```
Consecutive Numbers
```

ask yourself

```
Can I represent it as
```

```
Start + Length ?
```

If yes,

Write AP formula.

Convert the problem into an equation.

Instead of searching both variables,
iterate over the one having the smaller search space.

Recognition Checklist

- ✓ Consecutive numbers
 - ✓ Arithmetic progression
 - ✓ Need number of ways
 - ✓ Unknown start and length
 - ✓ Derive equation
 - ✓ Iterate over smaller variable
-

Similar Interview Problems

1. Consecutive Numbers Sum (Leetcode 829)
 2. Count ways to express N as consecutive integers
 3. Find all consecutive sequences with given sum
 4. Continuous Subarray Sum (different pattern—Prefix Sum)
 5. Arithmetic Progression based sequence problems
-

Common Mistakes

- ✗ Iterating over every starting number
 - ✗ Forgetting that sequence length starts from 2
 - ✗ Forgetting $x > 0$
 - ✗ Forgetting $x \% 2 == 0$
 - ✗ Looping till n instead of $\sqrt{2n}$
 - ✗ Using floating-point division instead of integer arithmetic
-

Real-World Pattern

Many optimization problems have **two unknown variables**.

$$\text{Cost} = \text{Price} \times \text{Quantity}$$
$$\text{Area} = \text{Length} \times \text{Breadth}$$
$$\text{AP Sum} = \text{Start} + \text{Length}$$

Instead of brute-forcing both variables,

1. Build the equation.
2. Rearrange it.
3. Compute one variable from the other.
4. Iterate over the variable with the **smaller search space**.

This is a very common interview optimization technique.

Permanent DSA Sheet

Recognition

```
Consecutive Numbers
  ↓
Arithmetic Progression
  ↓
Represent as (start,length)
  ↓
Write AP Formula
  ↓
Set equal to target
  ↓
Solve for one variable
  ↓
Choose smaller search space
  ↓
Validate mathematical conditions
```

Questions to Ask Yourself

1. Is this an AP?
2. What are the unknowns?
3. Can I derive an equation?

4. Which variable has fewer possibilities?
5. Can I calculate one variable directly?
6. What mathematical conditions make the answer valid?

Biggest Lesson Today

Optimization did not come from the AP formula itself. It came from transforming the AP equation into a search over a single variable (k), because k has a much smaller search space ($\sqrt{2n}$) than the starting value (a).

Pattern Name (Save in DSA Notebook)

```
Primary
★★★★★ Mathematical Derivation

Secondary
★★★★★ Equation Transformation

Optimization
★★★★★ Search Space Reduction

Supporting
★★★★☆ Arithmetic Progression

Supporting
★★★★☆ Integer Validation

Supporting
★★★★☆ Boundary Derivation using Mathematics
```

Where would I keep this in my DSA Notebook?

I would create a separate section:

```
Mathematical Optimization
|
├─ Number Theory
├─ Arithmetic Progression
├─ Equation Derivation
├─ Iterate Smaller Variable
├─ Algebraic Search Space Reduction
└─ Integer Validation
```



Similar Problems Under This Pattern

Problem	Pattern
Consecutive Numbers Sum (LC 829)	AP + Mathematical Derivation
Reach a Number (LC 754)	Mathematical Derivation
Poor Pigs (LC 458)	Mathematical Reasoning
Mirror Reflection (LC 858)	GCD / Math
Water & Jug Problem (LC 365)	Number Theory + GCD
Sum of Consecutive Numbers	AP + Equation Derivation

Permanent Pattern Sheet (Add this to your notebook)

Pattern Name:

Mathematical Derivation / Equation Transformation



Recognition:

Question has

- Unknown variables
- Formula already exists
- Brute force is huge
- Need count/minimum/maximum
- Mathematical relationship

Thinking Framework

Step 1

Represent using variables.

Step 2

Write mathematical equation.

Step 3

Rearrange equation.

Step 4

Choose the smaller search variable.

Step 5

Validate constraints

(Integer, Positive, Divisible, etc.)

Step 6

Implement.